Amitash Nanda

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EDUCATION

- **University of California San Diego** — **San Diego, California**
  Master of Science in Electrical and Computer Engineering — *Expected Graduation June 2023*
  - Fall Coursework: Statistical Learning, Principles of AI Probabilistic Reasoning and Decision-Making
- **College of Engineering and Technology** — **Bhubaneswar, India**
  Bachelor of Technology in Instrumentation and Electronics Engineering, GPA - 8.60/10.0 — *Aug 2015 - May 2019*
  - Relevant Coursework: Embedded Systems, Data Structure, Soft Computing, Control System

TECHNICAL SKILLS

- **Programming Languages:** *Python, C/C++, HTML, CSS, JavaScript, MATLAB*
- **Libraries:** *PyTorch, TensorFlow, Scikit-learn, Pandas, OpenCV, Numpy, Matplotlib, Keras, Flask, NetworkX, Bokeh*
- **Frameworks, and Simulator:** *Jupyter, PyQt, Tkinter, Gazebo, V-Rep, RViz, Node.js, Express.js, 3D Experience*
- **Database, Tools, and Cloud:** *SQL, MongoDB, LaTeX Git, Jetson, ROS, AWS S3, AWS RoboMaker, Azure, Bitbucket*

PROFESSIONAL EXPERIENCE

- **University of California San Diego - Boolean Lab** — **San Diego, California**
  Graduate Student Researcher — *Fall 2021*
  - Working on implementing boolean implications and analyzing biological datasets using Boolean Network Explorer.
  - Investigating different methodologies to develop a deep-learning based organoid mining system.
- **Salk Institute of Biological Studies, San Diego - Panda Lab** — **Remote**
  Research & Development Volunteer — *May 2021 - Aug 2021*
  - Worked on creating food networks and models with circadian stability under the supervision of Dr. Satchidananda Panda.
  - Collected and processed the data from application and created interactive network visualization using NetworkX and Bokeh.
- **Accenture Technology - Application Engineering R&D** — **Bengaluru, India**
  Application Development Analyst — *Sept 2020 - Aug 2021*
  - Design and developed a novel chaotic testing application to test the robustness of a robot system.
  - Worked on the frontend GUI design, implemented Gazebo-ROS functionalities, and integrated on AWS S3 and RoboMaker.
- **Accenture Technology - Industry X Services** — **Bengaluru, India**
  Application Development Associate — *Jun 2019 - Aug 2020*
  - Automated the end-to-end user stories of the Airbus Product Life Cycle Management using Image and Text-based recognition.
  - Worked on customization and integration of PLM functionalities using RESTful web services and designed a hybrid mobile application using Apache Cordova.

PROJECTS

- **Chaos Robo Application - Create chaos to avoid chaos**
  Chaos Engineering implementation to develop a fault tolerant robot application — *Aug 2020 - Aug 2021*
  - Designed and developed perturbation, node kill, add objects, and latency functionalities for robot chaos testing application.
- **Plant Sustainability Enhancement using Deep Learning Techniques**
  Efficient ResNet-CNN model for health classification of plant saplings — *Apr 2019 - Jul 2020*
  - Designed and Developed an autonomous, robust, and non-contact system for the saplings to monitor their growth, vicinity, and health conditions using ResNet-CNN and MQTT IoT protocol.

PUBLICATIONS

- A Nanda, et al.,"Computer Vision Assisted Autonomous Intra Row Weeder", ICIT (IEEE), 2018
- A Nanda, et al.,"NeuCASL: From Logic Design to System Simulation of Neuromorphic Engines", FMCAD, 2021 (accepted)